

Ex: No: 6A
Date: 15/04/26

Customer Purchase Category Prediction Using Naive Bayes Classifier

Problem Description:

An e-commerce company wants to predict whether a customer will make a **High** or **Low** purchase based on demographic and behavioral attributes. The dataset contains both **categorical** (Gender, Membership) and **numerical** (Age, Visits per Month) features.

To perform probabilistic classification and handle mixed data types efficiently, the **Naive Bayes classifier** is used.

Objective:

To implement the Naive Bayes classification algorithm in R by training a model on customer data, predicting purchase categories, and evaluating classification accuracy.

Procedure:

- Step 1: Create Dataset - customer_data

	Age	Visits	Gender	Membership	Purchase
1	22	2	Male	Regular	Low
2	35	5	Female	Gold	High
3	40	6	Female	Gold	High
4	28	3	Male	Regular	Low
5	50	8	Female	Gold	High
6	45	7	Male	Silver	High
7	30	4	Female	Silver	Medium
8	55	9	Male	Gold	High
9	27	3	Male	Regular	Low
10	38	6	Female	Silver	Medium

customer_data <- data.frame (

Age = c(22, 35, 40, 28, 50, 45, 30, 55, 27, 38) ,

Visits = c(2, 5, 6, 3, 8, 7, 4, 9, 3, 6) ,

Gender = c("Male", "Female", "Female", "Male", "Female", "Male",
"Female", "Male", "Male", "Female") ,

Membership = c("Regular", "Gold", "Gold", "Regular", "Gold", "Silver",
"Silver", "Gold", "Regular", "Silver") ,

Purchase = c("Low", "High", "High", "Low", "High", "High", "Medium",
"High", "Low", "Medium"))

Naive Bayes Classifier for discrete predictors

call:

naive Bayes : default $k=y$, $y=y$ Laplace = Laplace

Aprior probabilities:

High	Low	medium
0.5714286	0.2857143	0.1428571

Conditional probabilities:

	Age		Gender	female	male
	[17]	[27]			
High	42.75	8.539128	High	0.5	0.5
Low	24.50	3.535534	Low	0.0	1.0
medium	38.00	NA	medium	1.0	0.0

membership

	High	Low	medium	Gold	Ruby	Silver
High	0.75	0.00	0.25	0.75	0.00	0.25
Low	0.00	1.00	0.00	0.00	1.00	0.00
medium	0.00	0.00	1.00	0.00	0.00	1.00

- Step 2: Convert Categorical Variables to Factors

```
customer_data$Gender <- as.factor(customer_data$Gender)
```

```
customer_data$membership <- as.factor(customer_data$membership)
```

```
customer_data$Purchase <- as.factor(customer_data$Purchase)
```

- Step 3: Train(70%)–Test Split(30%) use sample()

```
set.seed
```

```
train_index <- sample(1:nrow(customer_data), 0.7 * nrow(customer_data))
```

```
train_data <- customer_data[train_index,]
```

```
test_data <- customer_data[-train_index,]
```

- Step 4: Apply Naive Bayes Algorithm

```
#install packages "e1071" and load
```

```
install.packages("e1071")
```

```
library(e1071)
```

```
model <- naiveBayes(Purchase ~., data=train_data)
```

```
model
```

[1] High low Medium

Levels: High low medium

Actual

predicted

High

low

Medium

High

1

0

0

0

1

0

low

0

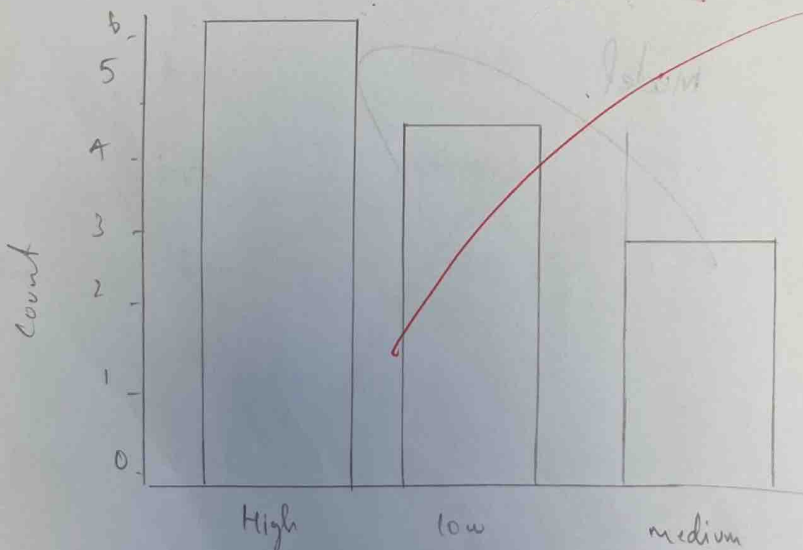
0

1

medium

"Accuracy: 0.666667"

Purchase category Distribution



- Step 5: Predict on Test Data

```
prediction <- predict(model, test.data)
```

```
print(prediction)
```

- Step 6: Confusion Matrix and Accuracy

```
confusion_mat <- table(predicted = predictions, Actual = test.data$purchase)
```

```
print(confusion_mat)
```

```
accuracy <- sum(diag(confusion_mat)) / sum(confusion_mat)
```

```
print(paste("Accuracy: ", accuracy))
```

- Step 7: Data Visualization for Naive Bayes Classification

- Class Distribution Plot Bar plot

- Shows how many customers belong to each purchase category.
- Helps identify class imbalance.

```
barplot(table(customer_data$purchase),
```

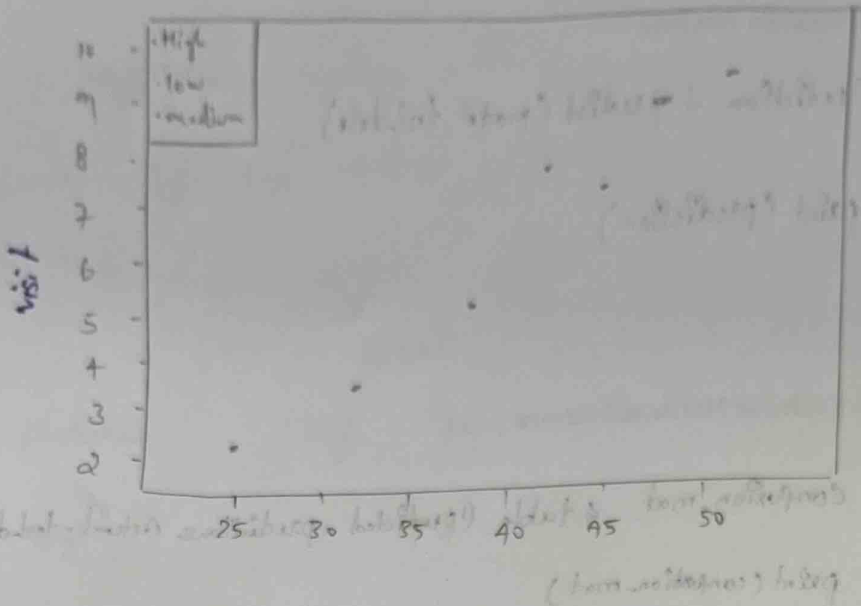
```
col = c("skyblue", "orange", "lightgreen")
```

```
main = "Purchase category Distribution",
```

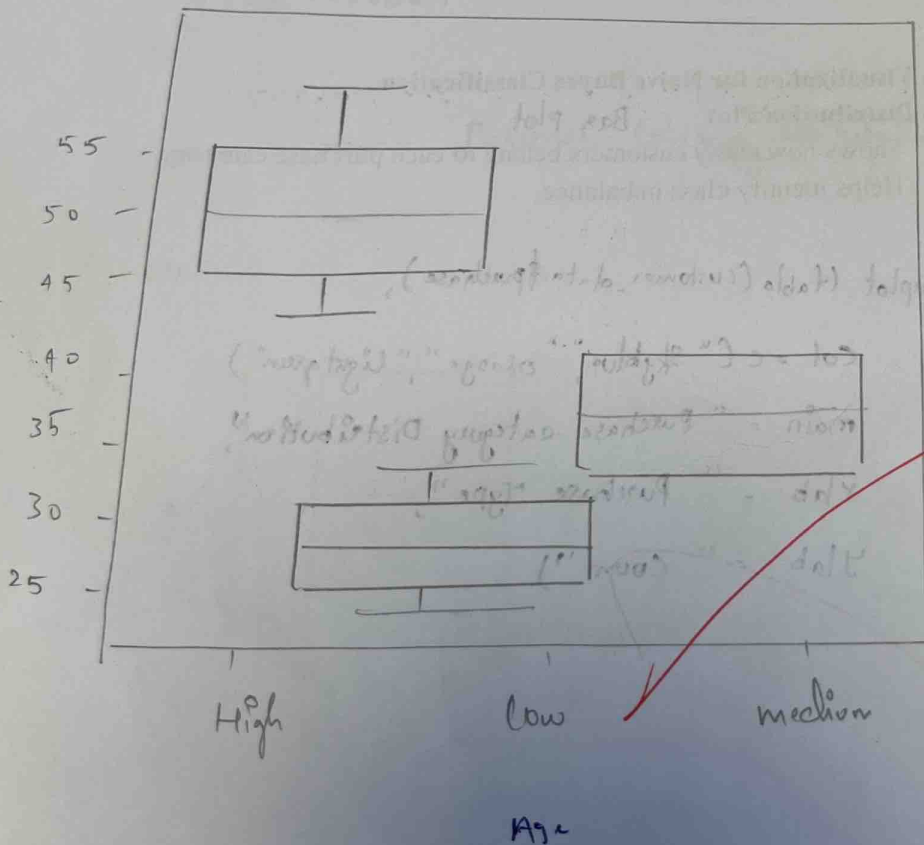
```
xlab = "Purchase Type",
```

```
ylab = "Count")
```


Age vs visit



Age Distribution by Purchase category



○ Scatter Plot (Age vs Visits)

- Displays customer grouping based on two numerical features – Age and Visits
- Colors represent purchase categories.

```
colors <- as.numeric(customer_data$purchase)
```

```
plot(customer_data$Age, customer_data$Visits,
```

```
col = colors,
```

```
pch = 19,
```

```
xlab = "Age", ylab = "Visits", main = "Age vs Visits")
```

```
legend("topleft",
```

```
legend = levels(customer_data$purchase),
```

```
col = 1:length(levels(customer_data$purchase)),
```

```
pch = 19)
```

○ Boxplot for Numerical Attributes - Age and Purchase Category

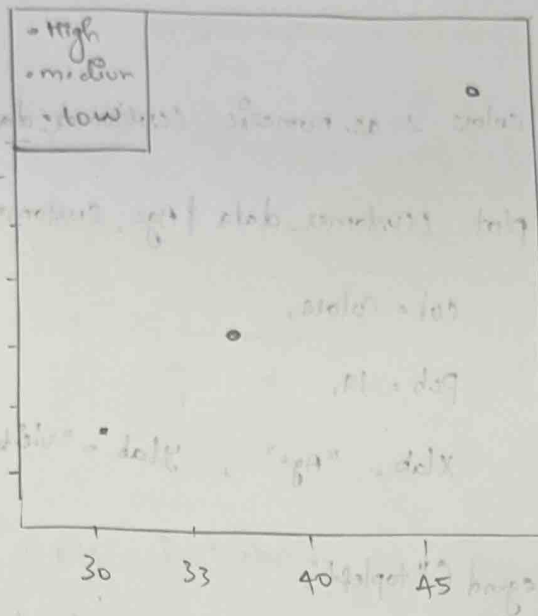
```
boxplot(Age ~ purchase,
```

```
data = customer_data,
```

```
col = "light green",
```

```
main = "Age Distribution by Purchase Category")
```

Prediction Results



○ Visualization of Prediction Results

```
plot (test_data$Age, test_data$Visits,
```

```
col = as.numeric (predictions),
```

```
pch = 19,
```

```
main = "prediction Results",
```

```
xlab = "Age",
```

```
ylab = "Visits")
```

```
legend ("topleft",
```

```
legend = levels (predictions),
```

```
col = 1:length (levels (predictions)),
```

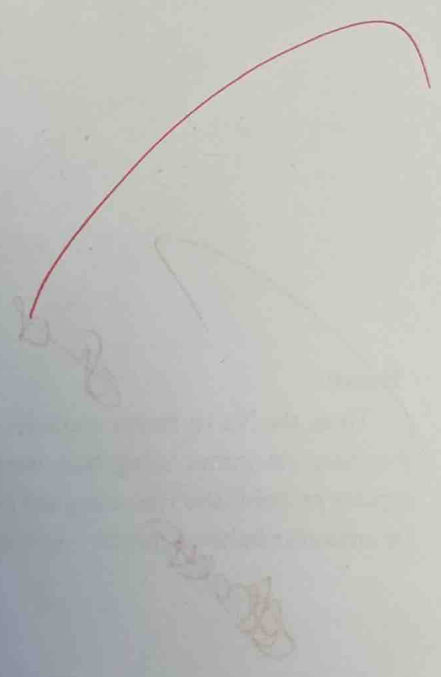
```
pch = 19)
```

Result:

Thus, the Naive Bayes classifier was successfully implemented in R to predict customer purchase categories using both numerical and categorical attributes. The model effectively applies probabilistic reasoning and produces accurate classification results, making it suitable for customer behavior prediction in e-commerce applications.

> head(iris)

	Sepal.length	Sepal.width	Petal.length	Petal.width	Species
1.	5.1	3.5	1.4	0.2	Setosa
2.	4.9	3.0	1.4	0.2	Setosa
3.	4.7	3.2	1.3	0.2	Setosa
4.	4.6	3.1	1.5	0.2	Setosa
5.	5.0	3.6	1.4	0.3	Setosa
6.	5.4	3.9	1.7	0.4	Setosa



Ex:No: 6B

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Iris Flower Species Classification Using Naive Bayes Algorithm

Problem Description:

The Iris dataset contains measurements of iris flowers belonging to three species: **Setosa**, **Versicolor**, and **Virginica**. Each flower is described using four numerical attributes: **Sepal Length**, **Sepal Width**, **Petal Length**, and **Petal Width**.

The objective is to classify iris flowers into their respective species using the **Naive Bayes classifier** and visualize the data distribution and classification results.

Objective:

To implement the Naive Bayes classification algorithm in R for predicting iris flower species and to visualize the dataset and classification results.

Procedure:

- **Step 1: Load Required Libraries**

We load the necessary libraries for implementing the Naive Bayes algorithm.

```
# Install package
```

```
install.packages("e1071")
```

```
# Load required libraries
```

```
library(e1071)
```

- **Step 2: Load and Prepare the Dataset**

We use the iris dataset for classification and check its structure.

```
# Load the iris dataset
```

```
data(iris)
```

```
# Display first few rows
```

```
head(iris)
```

Naive Bayes Classifier for Discrete Probability:

Call:

Naive Bayes (formula = Species, data = train data)

A-priori probabilities

	Setosa	Versicolour	Virginica
	0.3238015	0.328015	0.5528810

Conditional probabilities

Sepal. length:

	Setosa	Versicolour	Virginica
mean	5.006383	5.936170	0.542857
sd	0.3524896	0.465710	0.562552

Sepal. width:

	Setosa	Versicolour	Virginica
mean	3.428571	2.762766	0.542857
sd	3.43571	1.869105	0.112857

Petal. length

	Setosa	Versicolour	Virginica
mean	1.46428	4.362128	5.551020
sd	0.1728656	0.469915	0.535755

Petal. width

	Setosa	Versicolour	Virginica
Mean	0.242857	1.542553	2.028571
sd	0.106972	0.996773	0.2746504

- **Step 3: Split the Dataset into Training and Testing Sets**

We split the dataset into 70% training and 30% testing. `create()`

```
Set.seed(123)
```

```
train_index <- createDataPartition(cbind(species, p = 0.7), list = FALSE)
```

```
train_data <- iris[train_index, ]
```

```
test_data <- iris[-train_index, ]
```

- **Step 4: Train the Naïve Bayes Model**

We build the Naïve Bayes model using the `naiveBayes()` function from the `e1071` package.

```
# Train Naïve Bayes classifier
```

```
model <- naiveBayes(species ~., data = train_data)
```

```
# Print model details
```

```
print(model)
```

- **Step 5: Make Predictions on Test Data**

We use the trained model to make predictions on the test dataset.

```
# Make predictions
```

```
predictions <- predict(model, test_data)
```

```
# Display predictions
```

```
head(predictions)
```

Actual (prediction)

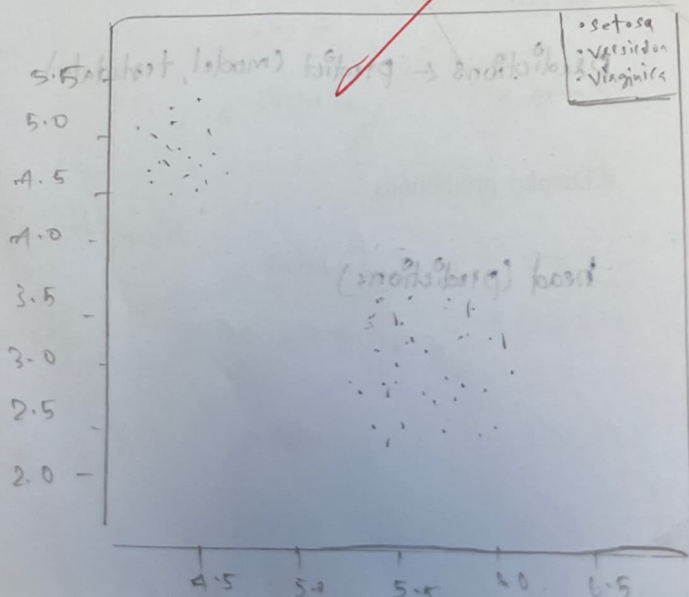
[1]	setosa	virginica	setosa	versicolor	virginica	versicolor
[2]	setosa	versicolor	virginica	setosa	setosa	versicolor
[22]	versicolor	setosa	virginica	versicolor	setosa	setosa
[23]	versicolor	virginica	setosa	versicolor	setosa	versicolor
[43]	virginica	setosa	setosa	setosa	versicolor	versicolor

Actual

predicted setosa versicolor virginica

Setosa	15	0	0
versicolor	0	14	1
virginica	0	1	14

[1] Accuracy : 0.95



- **Step 6: Evaluate Model Performance**

We compute accuracy using a confusion matrix.

Confusion matrix

`conf_matrix <- table(predicted = predictions, Actual = test_data$species)`

Print accuracy

`accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)`

- **Step 7: Visualize Classification Results**

We plot a scatter plot of Sepal.Length vs Sepal.Width with predicted class labels.

Add predictions to test data

`test_data$predicted_species <- predictions`

Scatter plot of Sepal.Length vs Sepal.Width

`ggplot(test_data, aes(x = Sepal.Length, y = Sepal.Width, color = predicted_species)) +
geom_point(size = 3) +
ggtitle("Naive Bayes Classification Results").`

Result:

Thus, the program successfully demonstrates the use of the Naive Bayes algorithm in R to classify iris flower species based on numerical attributes. The visualizations clearly show species separation, and the model achieves high classification accuracy, validating the effectiveness of Naive Bayes for multi-class classification problems.